Technology Stack

Date: 05 October 2026

Team ID: UNM-293 2026-2430

Project Name: FitBuddy-AI Fitness Plan Generator using Gemini Models

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Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML, CSS, JavaScript, Bootstrap	Used to create a simple, responsive and user-friendly interface for entering fitness details and displaying personalized plans.
2	**Backend / Server-Side**	Python with Flask	Provides lightweight server-side processing, handles user requests, connects the application with the Gemini model, and generates fitness-plan responses.
3	**Database / Data Storage**	SQLite	Stores user profile information, fitness preferences and generated plan data in a simple and lightweight database suitable for the project.
4	**Cloud / Hosting / Deployment**	Google Cloud / Firebase	Supports deployment of the application and provides a scalable environment for making FitBuddy accessible to users.
5	**Version Control & CI/CD**	GitHub	Enables source-code version control, collaboration, project tracking and reliable management of application updates.
6	**Third-Party APIs / Other Tools**	Google Gemini API	Provides AI capabilities to analyze user-provided fitness information and generate personalized fitness-plan recommendations.

Purpose: The selected technology stack is designed to support a maintainable, responsive and AI-powered FitBuddy application for generating personalized fitness plans.